Rotations, Reflections, and Root Systems

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Outline

1. Finite Reflection Groups
   - Dihedral Group
   - Reflections
2. Root Systems
   - Positive Root Systems
   - Simple Root Systems
3. Further Applications

What are groups?

Definition

A binary operation on a set G is a function $\star : G \times G \to G$. We say $\star$ is *associative* if for all $a, b, c \in G$, $a \star (b \star c) = (a \star b) \star c$.

Definition

A group is an ordered pair $(G, \star)$, where $\star$ is a binary operation defined on a set G, which satisfies

1. Associativity: $\star$ is associative.
2. Identity Element: there exists an $e \in G$ such that, for all $g \in G$, $g \star e = e \star g = g$.
3. Inverse Element: for all $g \in G$, there exists an element $g^{-1} \in G$ such that $g \star g^{-1} = g^{-1} \star g = e$.

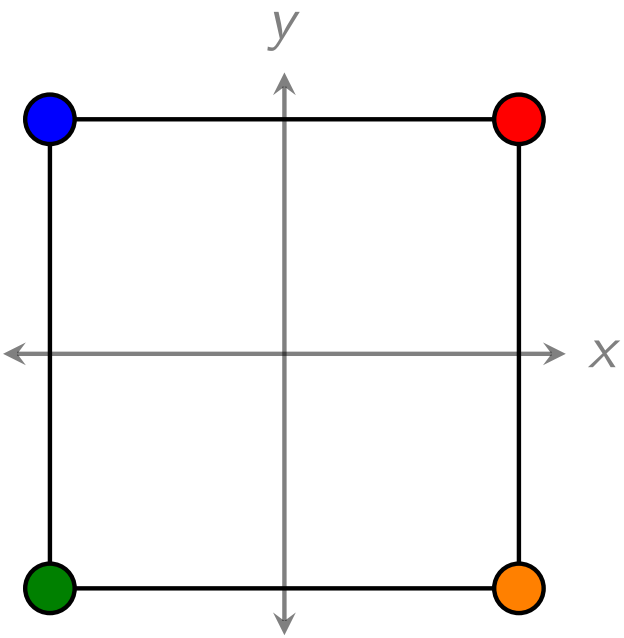
Dihedral group with rotations

Definition

The dihedral group of order $2n$, denoted D_{2n}, is the group of symmetries of a regular n-gon.

- Let r be a rotation by $\theta = \frac{2\pi}{n}$; let s be a reflection through a line of symmetry
- D_{2n} contains n reflections and n rotations
- $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$

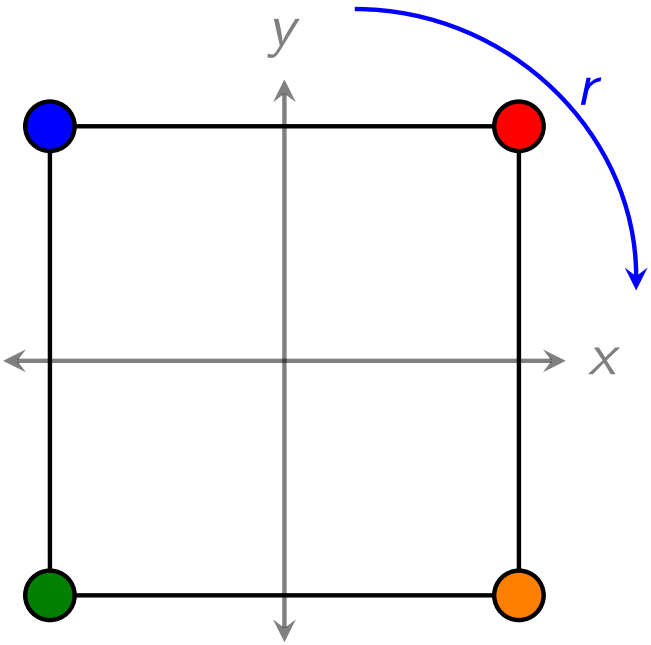
Concrete example with D_8



Example

- $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ is the symmetries of a square
- Let r be a rotation by 90° clockwise
- Let s be a reflection across the line $y = x$

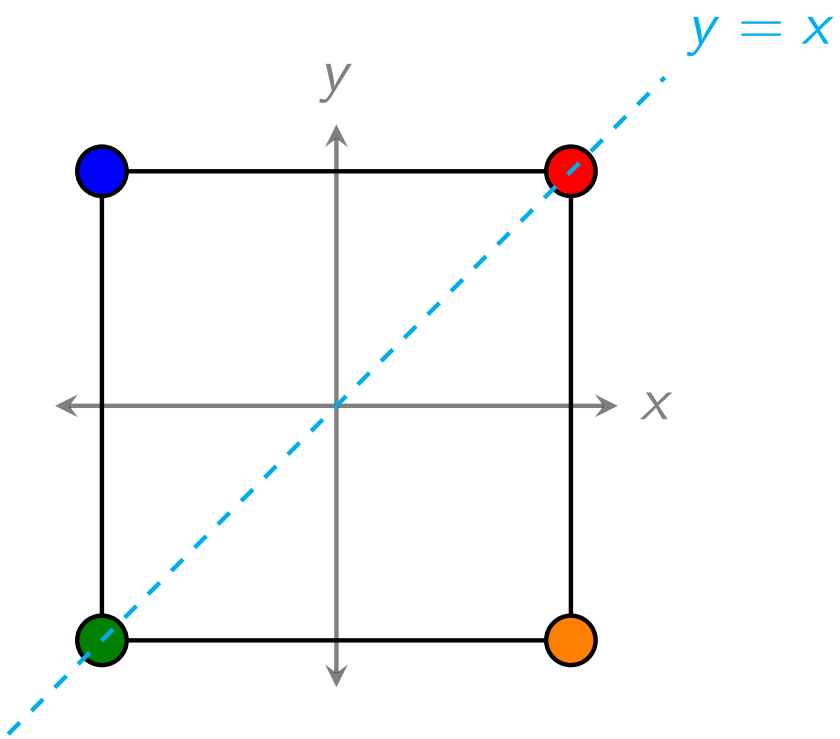
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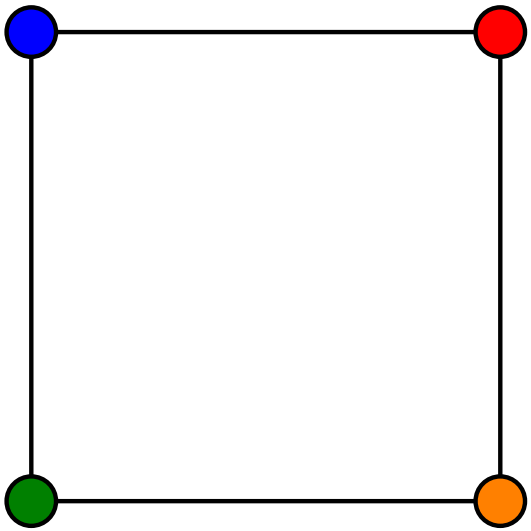
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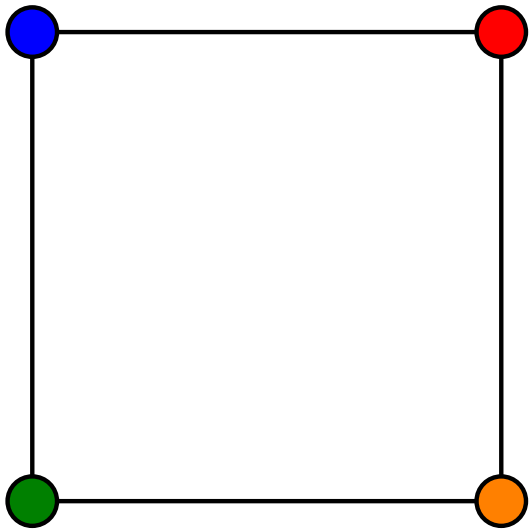
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Verify $rs = sr^{-1}$ in D_8

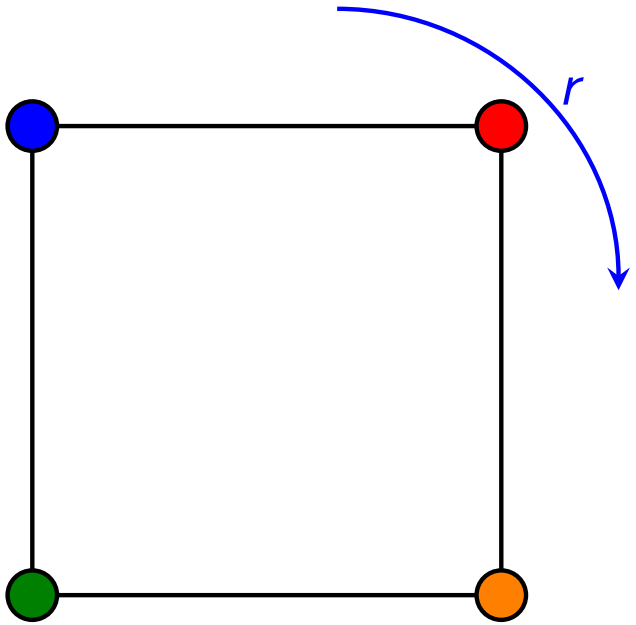


Identity

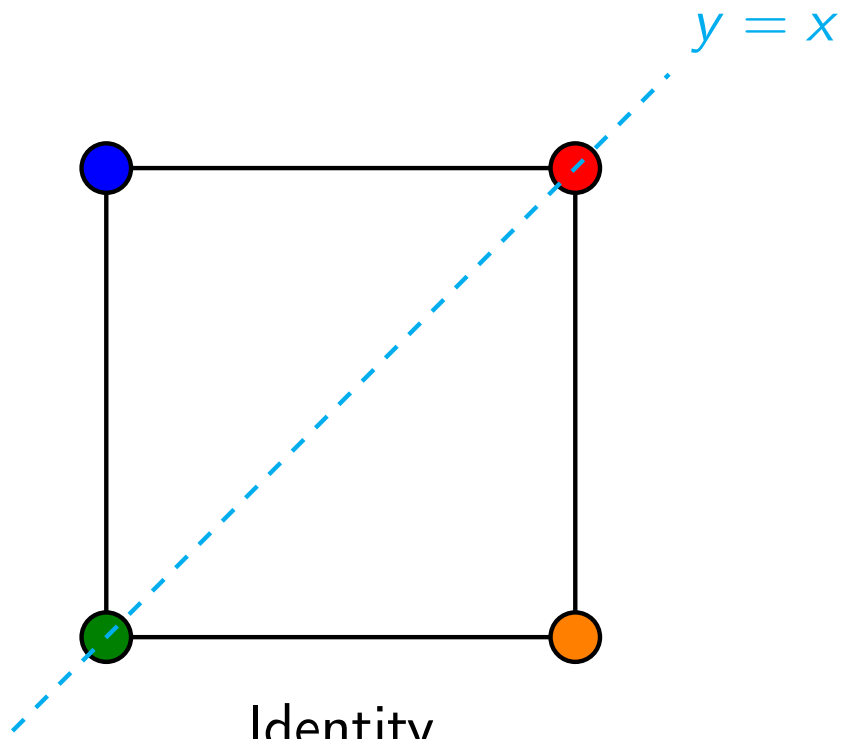


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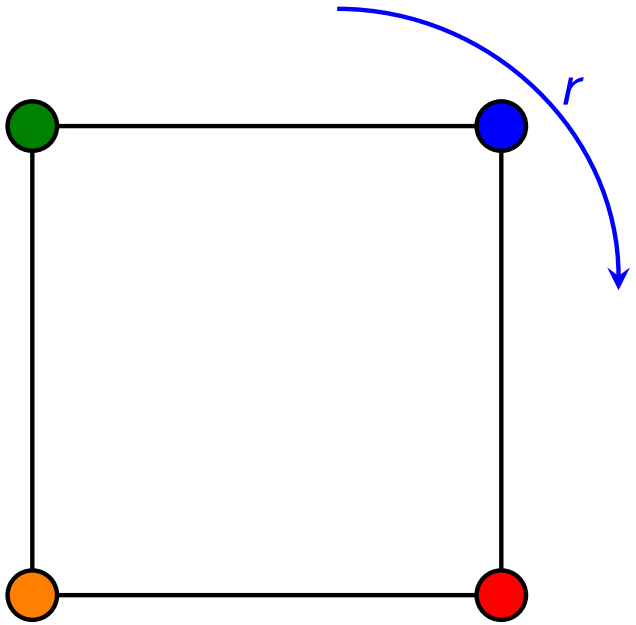


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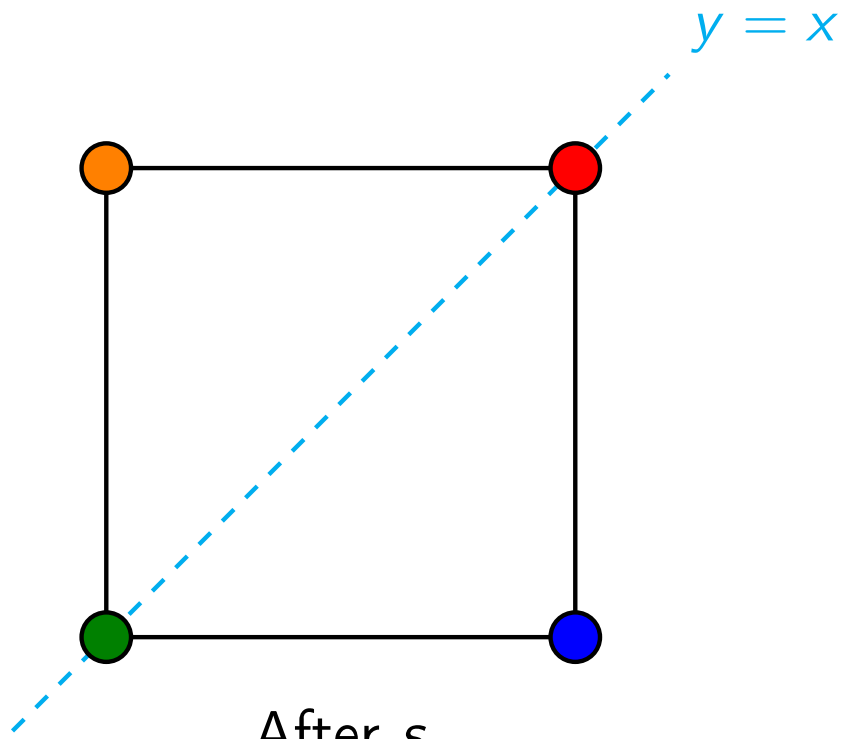


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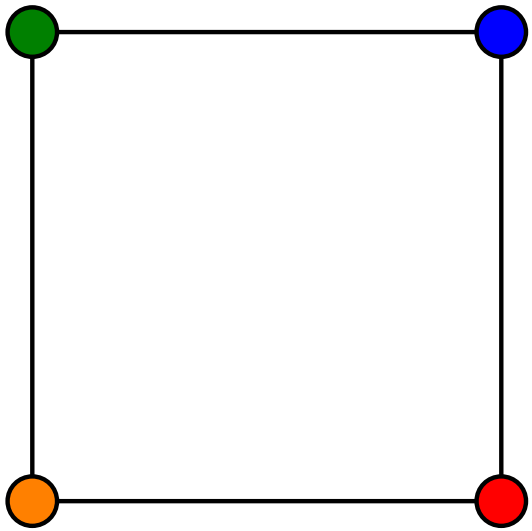


After r

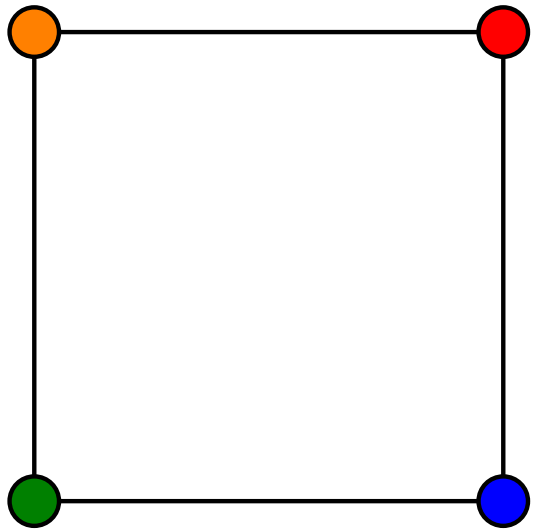


After s

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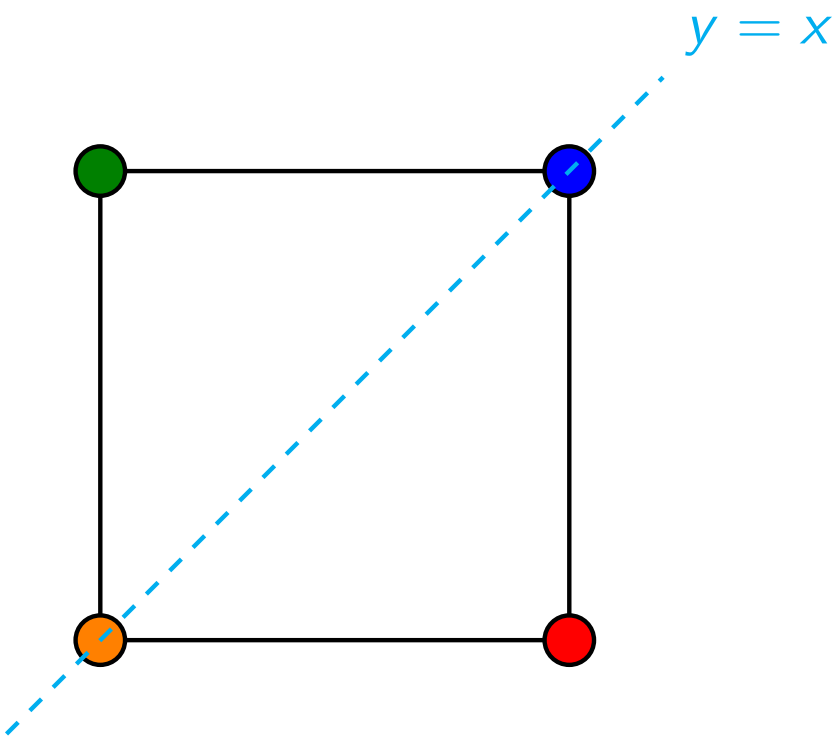


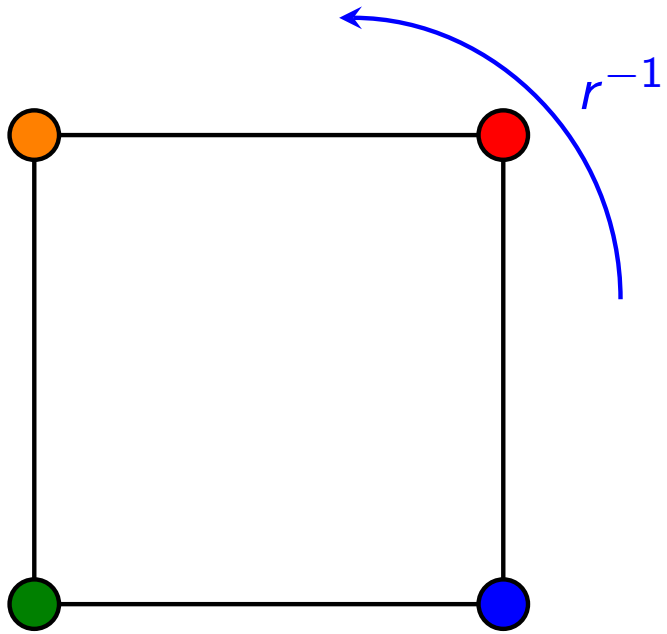
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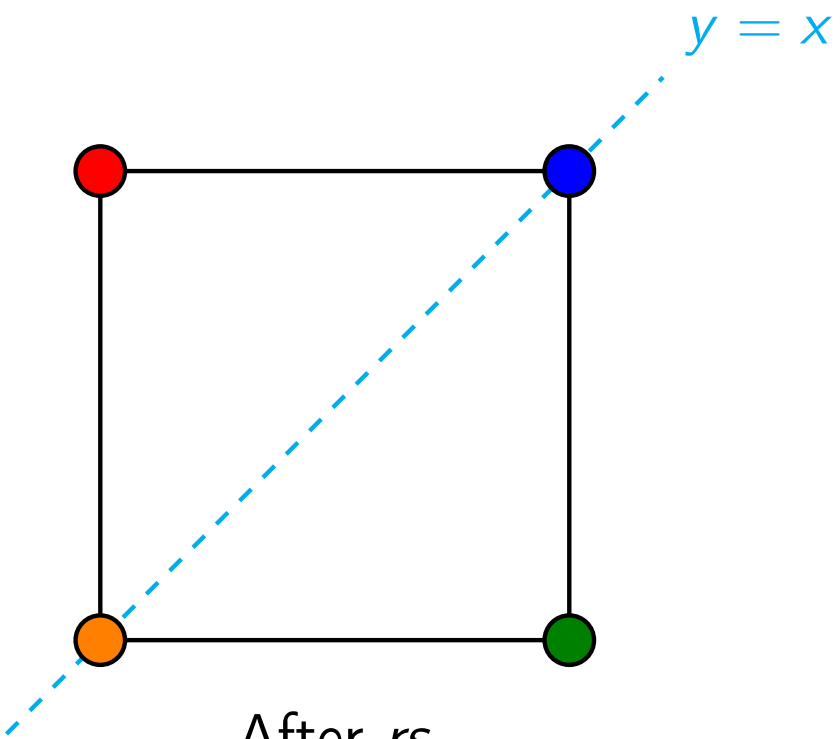
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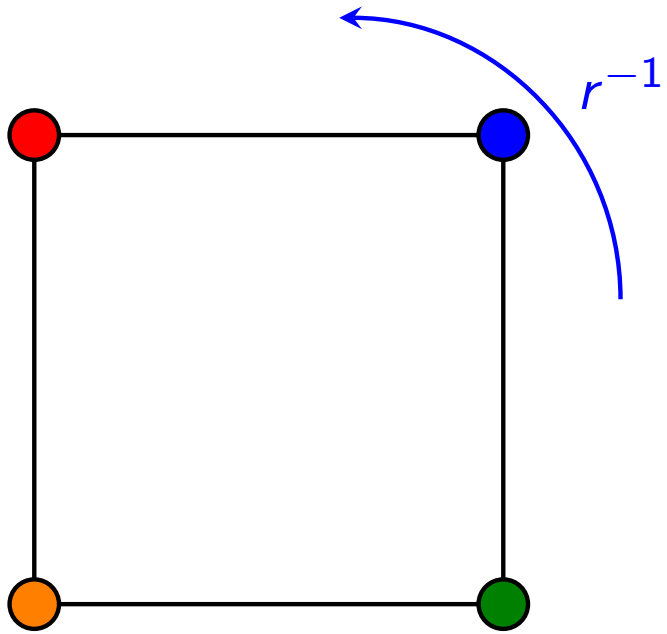




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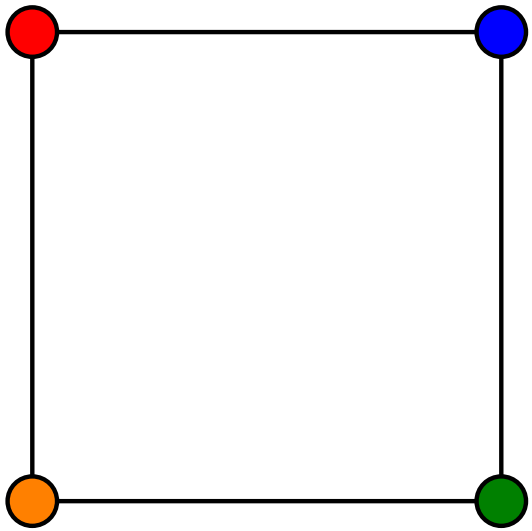


After rs

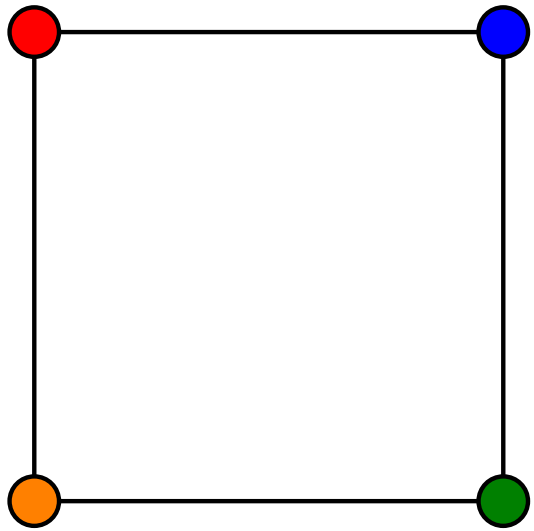


After sr^{-1}

Verify $rs = sr^{-1}$ in D_8



After rs



After sr^{-1}

A closer look at reflections

- Let V be a real euclidean space with an inner product $\langle \cdot, \cdot \rangle$.

Definition

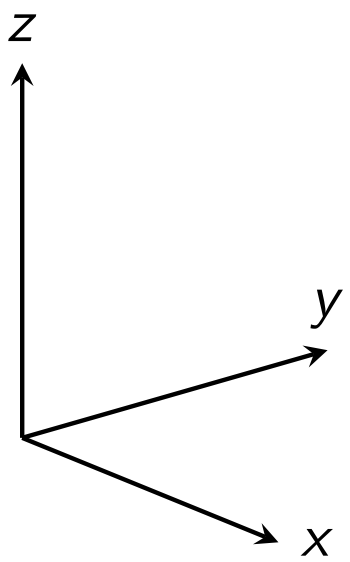
A reflection is a linear operator $s\colon V \to V$ which sends some nonzero vector α to its negative while fixing pointwise the hyperplane H_α orthogonal to α. In particular,

$$s_\alpha(\lambda) = \lambda - \frac{2\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle}\alpha.$$

Properties

1. $s_\alpha(\alpha) = -\alpha$ and $s_\alpha(\lambda) = \lambda$ for all $\lambda \in H_\alpha$
2. $\langle s_\alpha(\lambda), s_\alpha(\mu) \rangle = \langle \lambda, \mu \rangle$ for all $\alpha, \mu \in V$
3. $s_\alpha^2 = 1$

A closer look at reflections

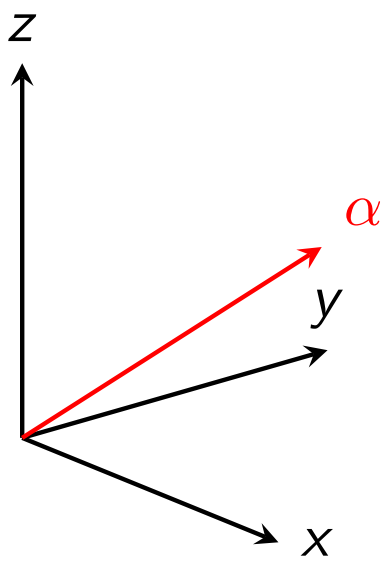


Example

Let $\alpha \neq 0 \in V$. Then, $H_\alpha = \{v \in V \mid \langle v, \alpha \rangle = 0\}$ and $\text{span}\{\alpha\} = \mathbb{R}\alpha$. For a given $\lambda \in V$, $\lambda = \text{proj}_{H_\alpha}\lambda + \text{proj}_{\mathbb{R}\alpha}\lambda$. Then,

$$\begin{aligned} s_\alpha(\lambda) &= s_\alpha(\text{proj}_{H_\alpha}\lambda) + s_\alpha(\text{proj}_{\mathbb{R}\alpha}\lambda) \\ &= \text{proj}_{H_\alpha}\lambda - \text{proj}_{\mathbb{R}\alpha}\lambda \\ &= (\lambda - \text{proj}_{\mathbb{R}\alpha}\lambda) - \text{proj}_{\mathbb{R}\alpha}\lambda \\ &= \lambda - 2\text{proj}_{\mathbb{R}\alpha}\lambda \\ &= \lambda - \frac{2\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle}\alpha. \end{aligned}$$

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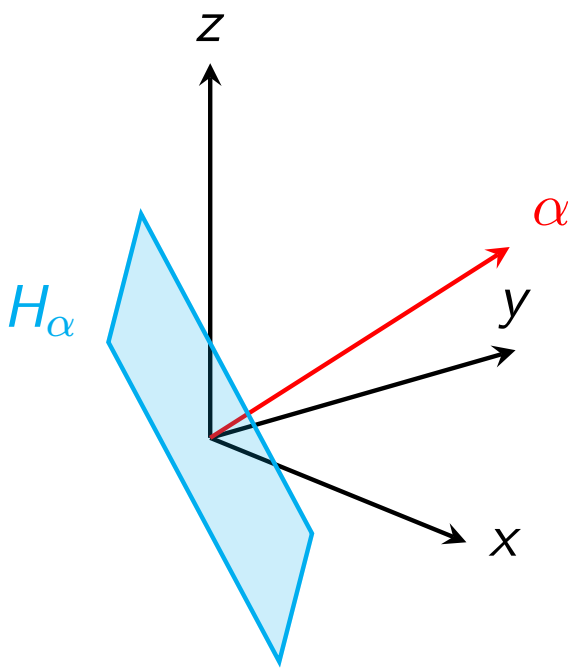


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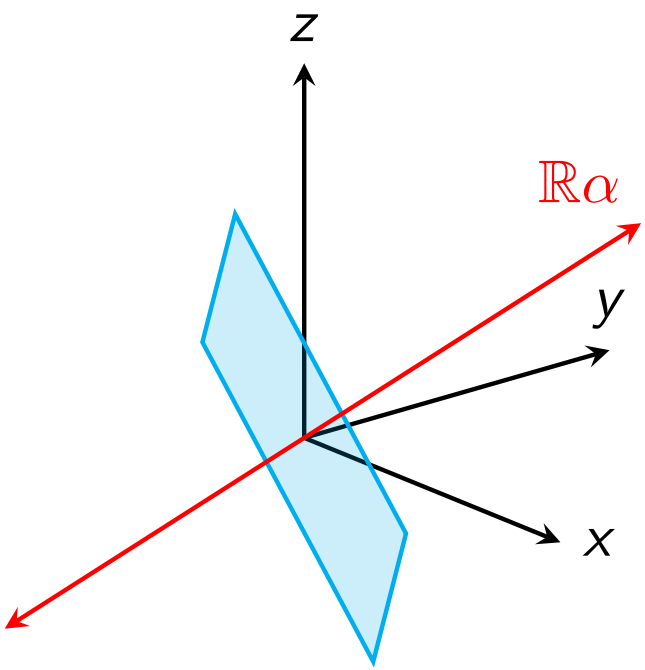


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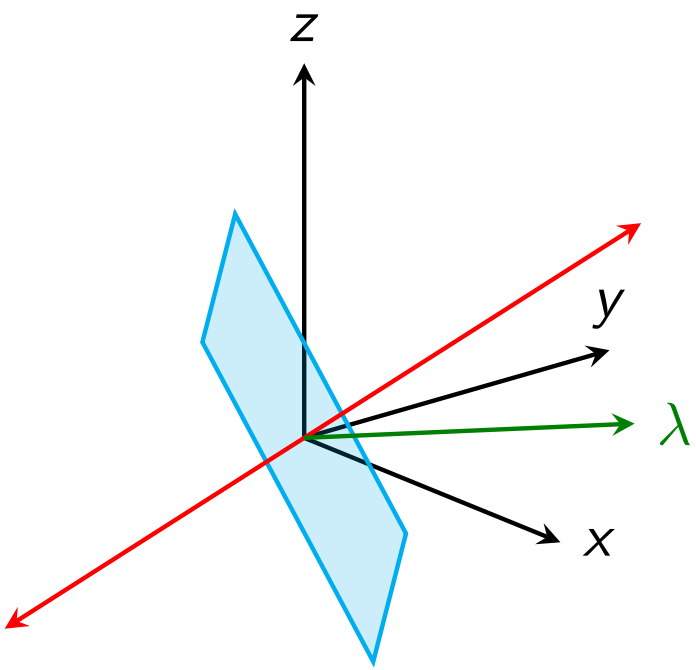


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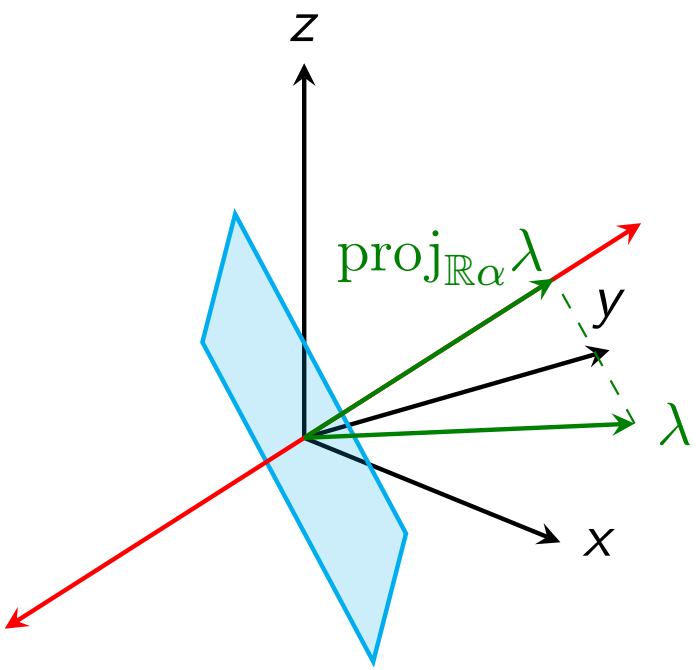


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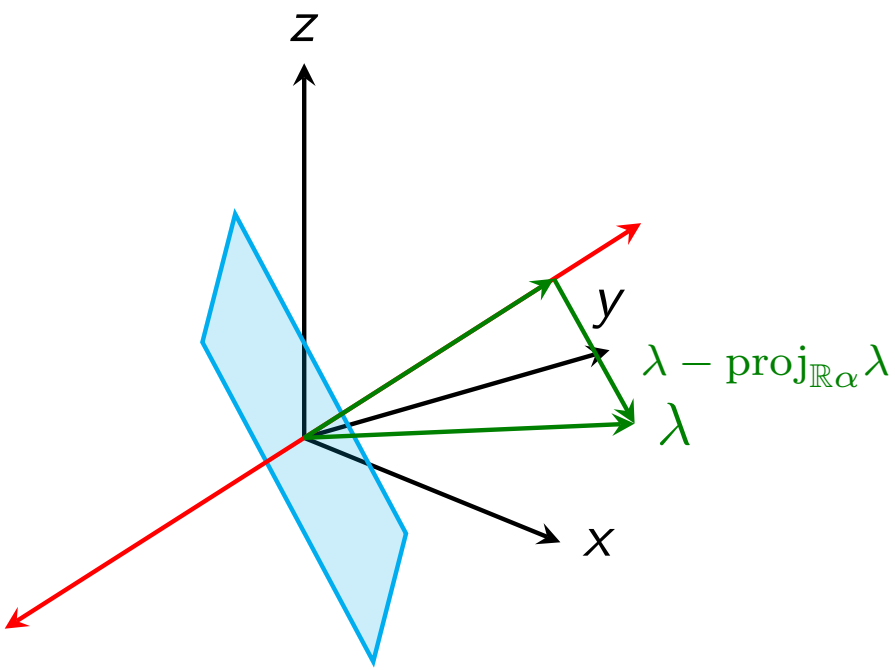


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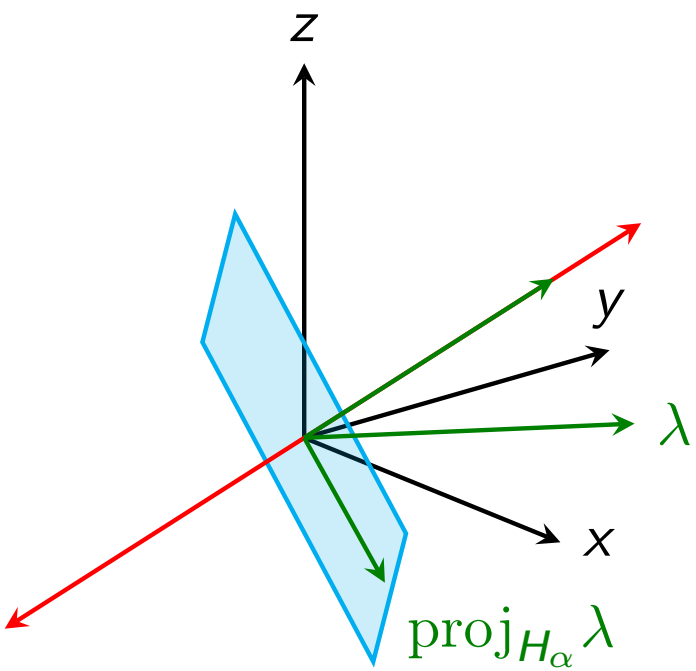


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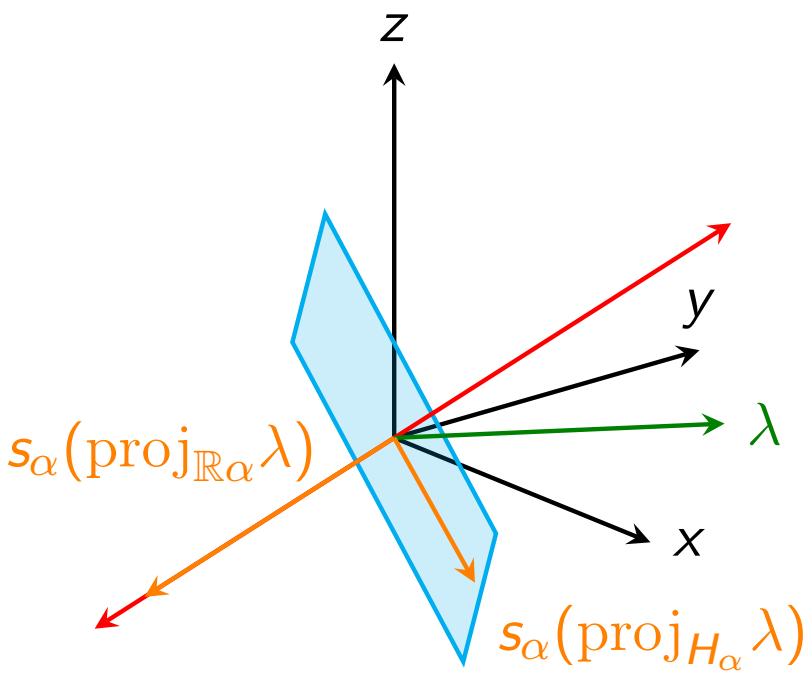


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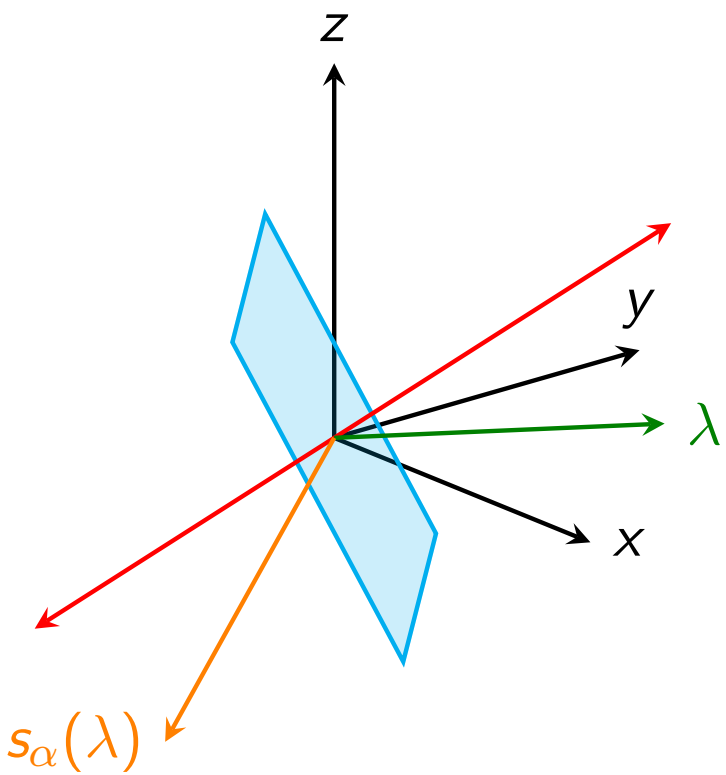


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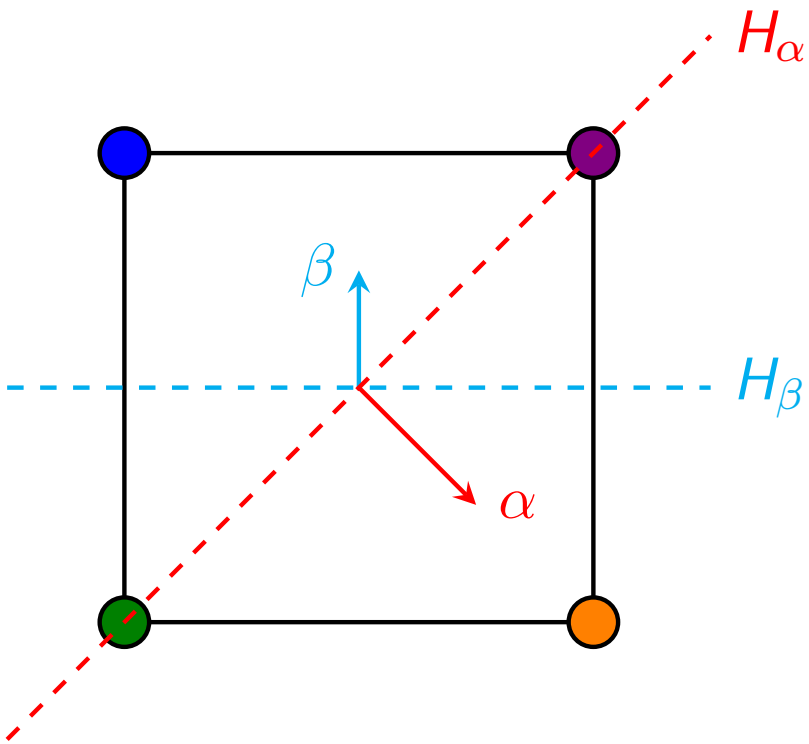
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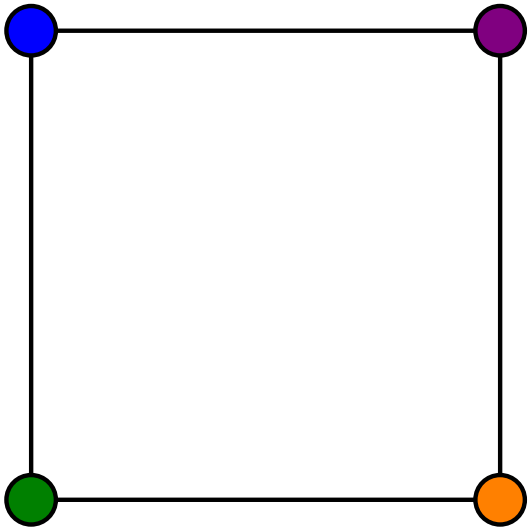
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Dihedral group with reflections

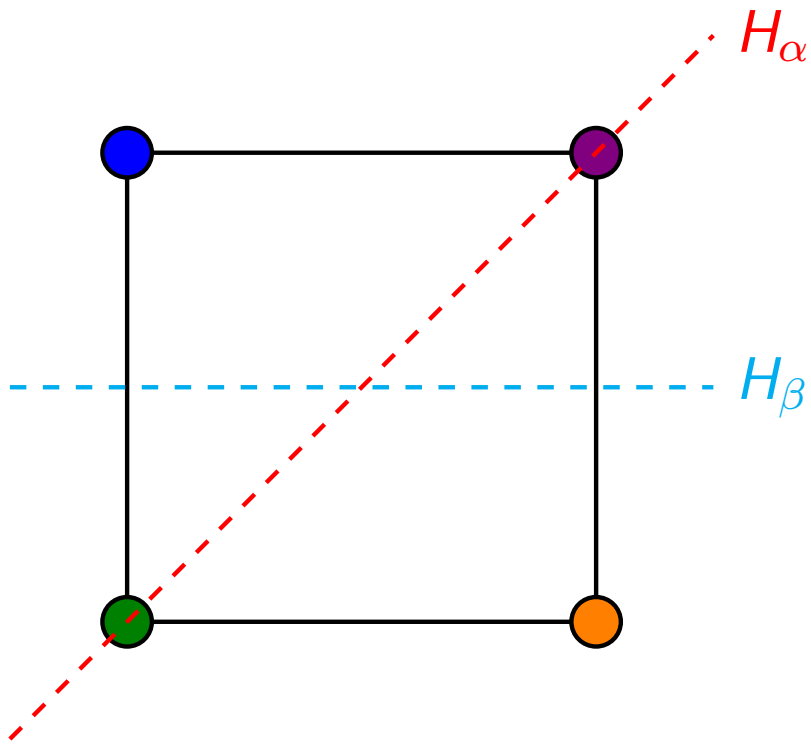
- Let H_α and H_β meet at an angle of $\theta = \frac{\pi}{n}$
- Then $s_\alpha s_\beta$ is a rotation through 2θ
- $D_{2n} = \langle s_\alpha, s_\beta \mid s_\alpha^2 = s_\beta^2 = (s_\alpha s_\beta)^n = 1 \rangle$



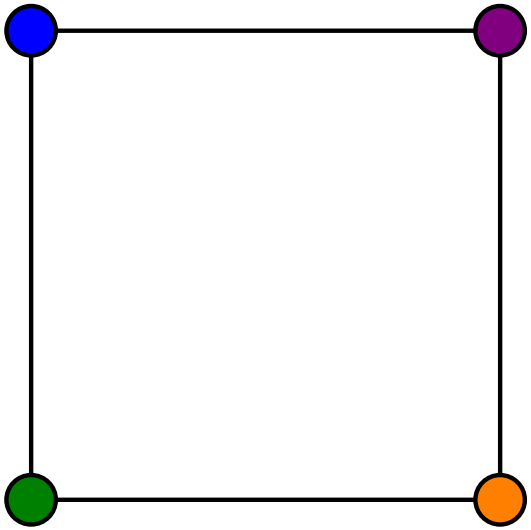
Verify $r = s_\alpha s_\beta$



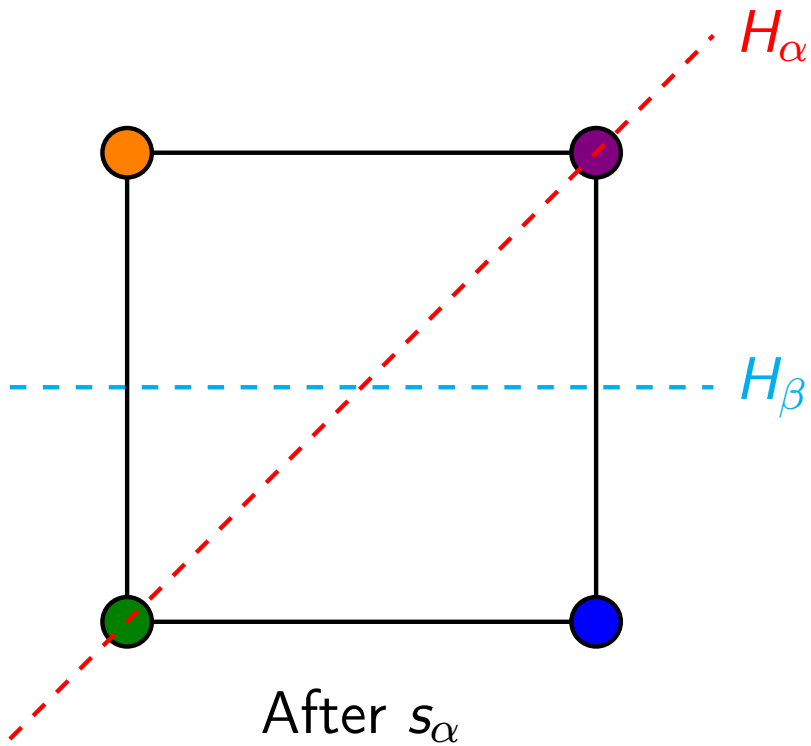
Identity



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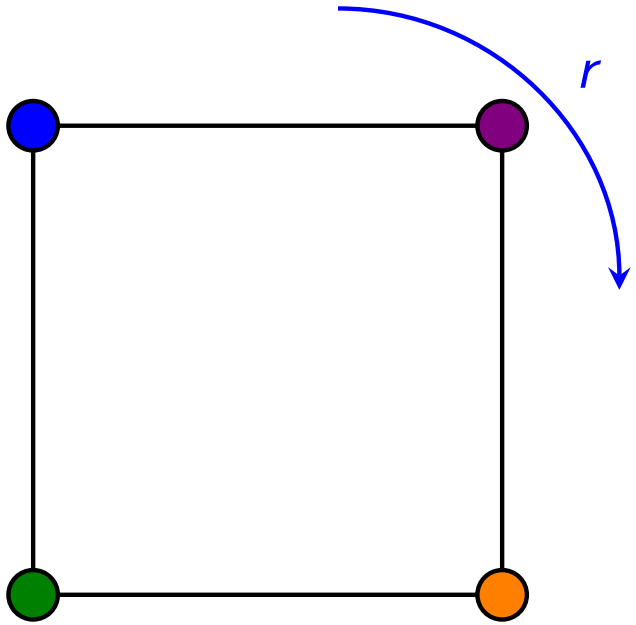


Identity

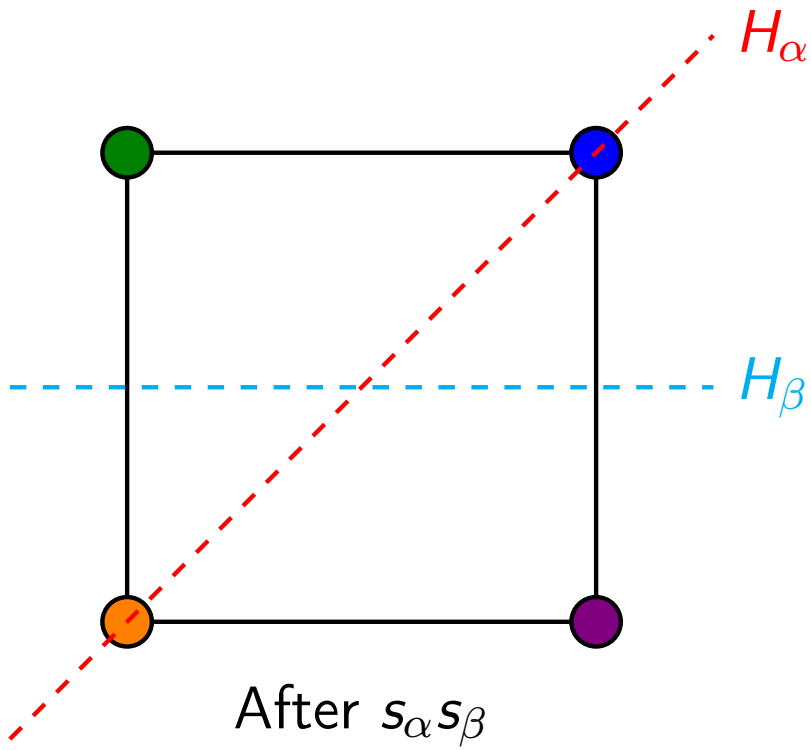


After s_α

Verify $r = s_\alpha s_\beta$



Identity



After $s_\alpha s_\beta$

What is a root system?

Definition

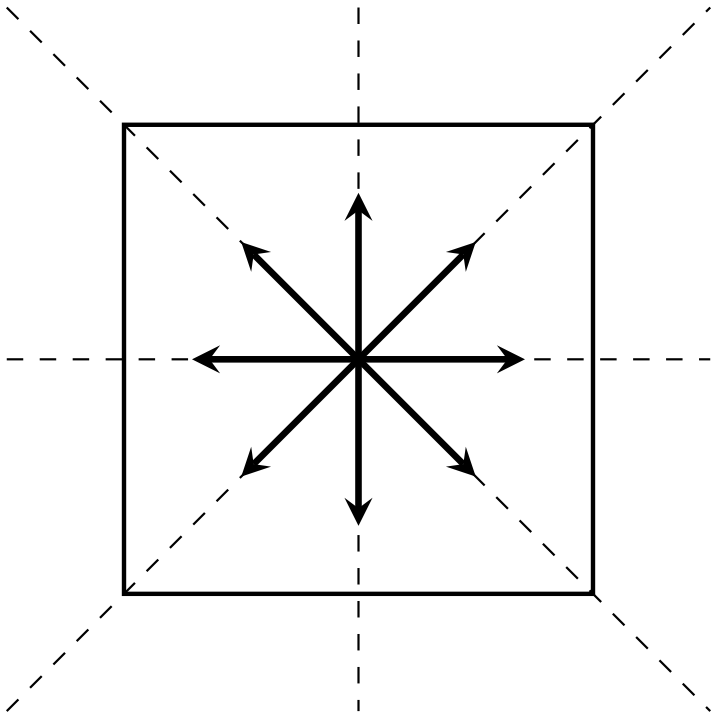
Let V be a vector space. Suppose we have a finite subset $\Phi \subset V$ which satisfies the following for any $\alpha \in \Phi$:

- $\alpha \neq 0$
- $\Phi \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$
- $s_\alpha(\Phi) = \Phi$

Let W be the reflection group generated by $\{s_\alpha : \alpha \in \Phi\}$. Then we say Φ is a root system for W.

Note that every finite reflection group has at least one (and possibly many) root system(s). For example, given a root system Φ, we can scale any element in Φ to obtain a new root system. Also, any reflection group which is induced by a root system is finite.

Root system of D_8



Total ordering

Definition

A total ordering on a vector space V is a relation $<$ which satisfies:

- For all $u, v \in V$, exactly one of $u < v$, $v < u$, $u = v$ holds.
- For all $u, v, w \in V$, if $u < v$ then $w + u < w + v$.
- Let $c \in \mathbb{R}$ and $u < v$. If $c < 0$ then $cu > cv$. If $c > 0$ then $cu < cv$.

We say $u \in V$ is *positive* if $u > 0$.

Note that we can construct a total ordering for any vector space V by choosing a basis for V and using a lexicographic ordering.

What is a positive system?

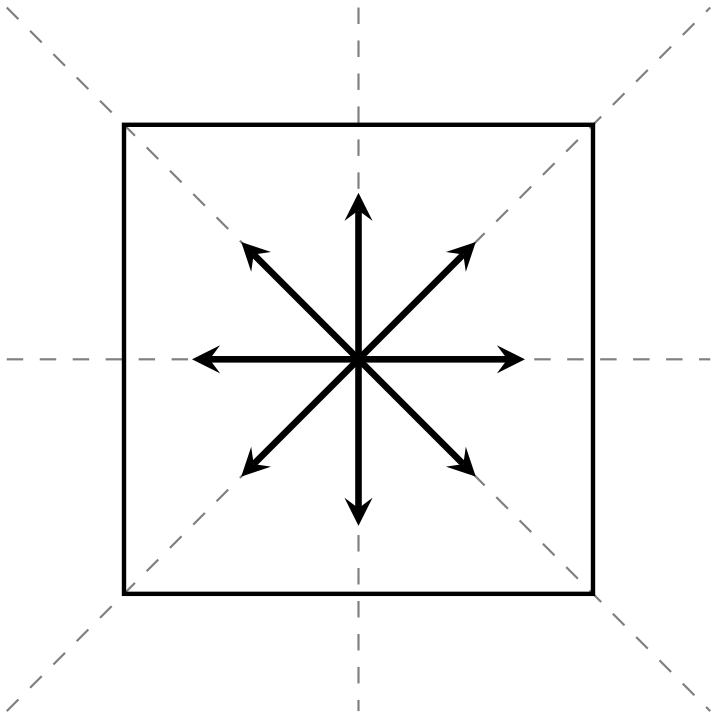
Definition

Fix a total ordering $<$ on a vector space V, and let Φ be a root system in V. Then $\Pi = \{r \in \Phi : r > 0\}$ is called a positive system.

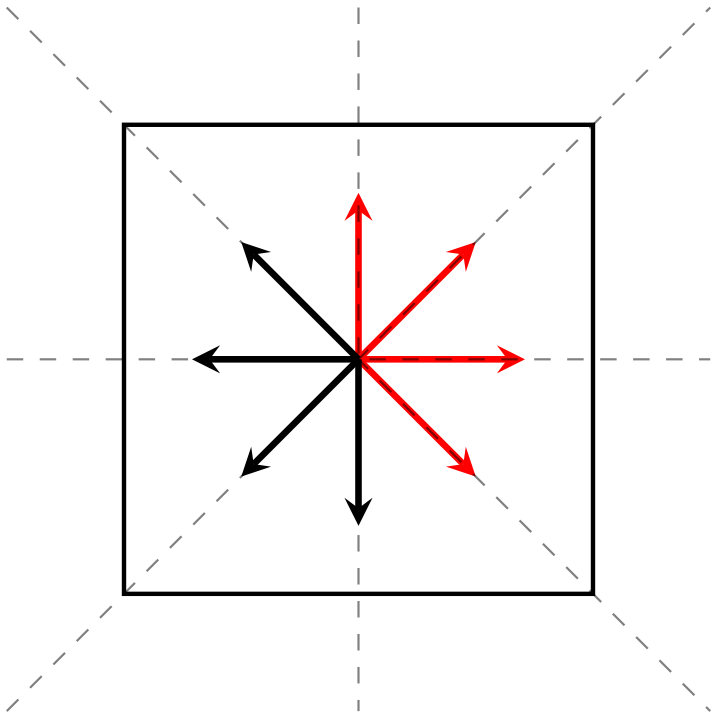
That is, a positive system is just the set of positive vectors in Φ under some total ordering. Negative systems are defined similarly.

Since roots come in pairs (like α and $-\alpha$) and roots cannot be zero, we see that any root system Φ can be partitioned into a positive and negative system.

Positive roots of D_8



Positive roots of D_8

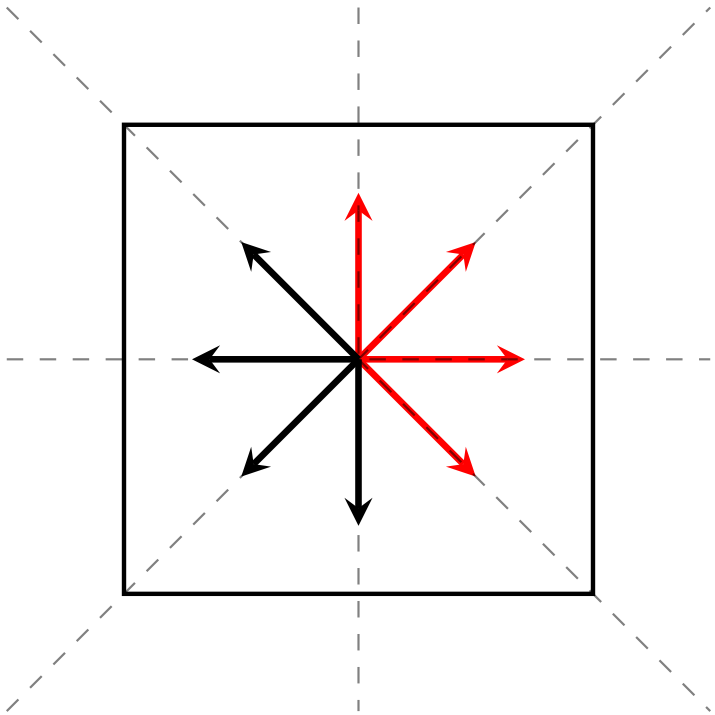


What is a simple system?

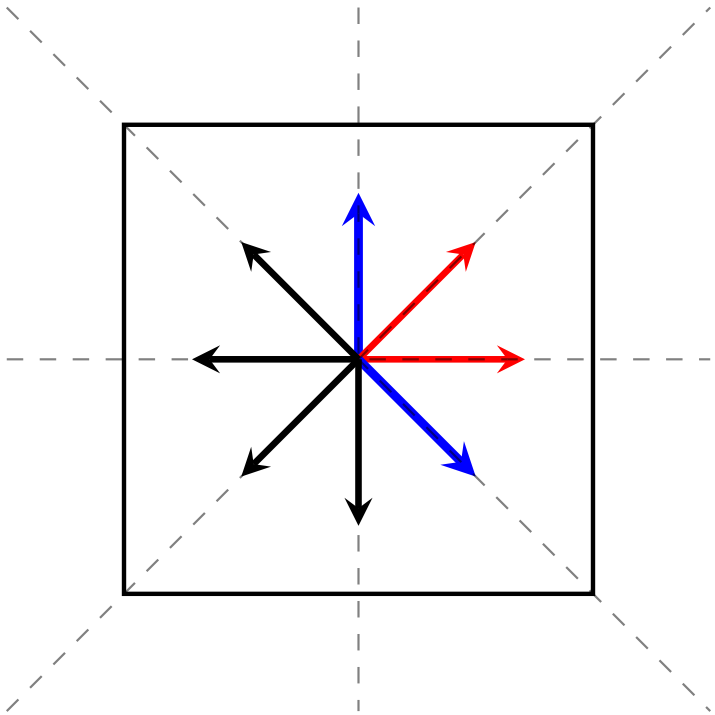
Definition

Let Φ be a root system in V. $\Delta \subset \Phi$ is a simple system if Δ is a basis for the $\mathbb{R}$-span of Φ, and every $\alpha \in \Phi$ is a linear combination of elements of Δ with coefficients all of the same sign.

Simple system of D_8



Simple system of D_8



Simple systems exist!

We'll now sketch the proof of the following theorem:

Theorem

1. *If $\Delta \subset \Phi$ is a simple system, then there is a unique positive system containing Δ.*
2. *Every positive system $\Pi \subset \Phi$ contains a unique simple system.*

By the definition of simple system, Δ is a basis for the $\mathbb{R}$-span of Φ, so $|\Delta| \leq \dim V$ and the elements of Δ are linearly independent. So we can extend Δ by adding some linearly independent vectors to form a new set Δ' which is a basis for V. Taking the lexicographic ordering with Δ', we get a positive system which contains Δ.

Suppose $\Delta \subset \Pi$ and $\Delta \subset \Pi'$. For any $p \in \Pi$, we know that p is a linear combination of elements of Δ with nonnegative coefficients, so $p \in \Pi'$. So $\Pi \subset \Pi'$. Using similar logic, we get $\Pi' \subset \Pi$, so $\Pi = \Pi'$. So the positive system containing Δ is indeed unique.

Choose Δ to be the minimal set such that each $p \in \Pi$ is a nonnegative linear combination of elements of Δ. Notice that a set with the specified properties always exists (we could choose $\Delta = \Pi$), so in particular the minimal set exists.

By construction, every $\alpha \in \Phi$ is a linear combination of Δ with coefficients of the same sign. This is because Π and $-\Pi$ partition Φ; if $\alpha \in \Pi$ then α is a nonnegative linear combination of Δ, and if $\alpha \in -\Pi$ then α is a nonpositive linear combination of Δ.

The previous slide also tells us that Δ spans the $\mathbb{R}$-span of Φ. So now we just need to show that Δ is linearly independent. We use the following lemma:

Lemma

For any $\alpha, \beta \in \Delta$, $\alpha \cdot \beta \leq 0$.

Suppose for sake of contradiction that $\alpha \cdot \beta > 0$. Then $s_\alpha(\beta) = \beta - c\alpha$, where $c > 0$. Then, condition on whether $s_\alpha(\beta)$ is in Π or $-\Pi$, and you can derive a contradiction in both cases.

Now that we believe the lemma, suppose for the sake of contradiction that Δ is not linearly independent. Then we have $\sum a_\alpha \alpha = 0$ with not all $a_\alpha = 0$. Discard all the zero coefficients and move the negative coefficients to the RHS to get $\sum b_\beta \beta = \sum c_\gamma \gamma$, where the sets of β and γ are disjoint and the coefficients are strictly positive.

Put $\sigma = \sum b_\beta \beta = \sum c_\gamma \gamma$. On one hand, we have $\sigma \cdot \sigma = |\sigma|^2 \geq 0$. But by the lemma, $\sigma \cdot \sigma = \sum b_\beta \beta \cdot \sum c_\gamma \gamma \leq 0$. So $\sigma \cdot \sigma = 0$, which means $\sigma = 0$. But σ is a positive linear combination of elements of Π, so $\sigma > 0$, and we have a contradiction. ■

Consequences of our theorem

From part 2 of the theorem, we now know that simple systems exist in any root system, since we can always find a positive system by inducing a total ordering. We now also know the following corollary, which is useful for finding real examples of simple systems:

Corollary

If $\Delta \subset \Phi$ is a simple system, then $\alpha \cdot \beta \leq 0$ for all $\alpha \neq \beta$ in Δ.

Applications of root systems

- Each root system determines a Weyl group, which is a kind of Coxeter group.
- Irreducible root systems are used to classify simple Lie algebras.

Summary

- A reflection group is generated by reflections in a vector space; think of the dihedral group as an example.
- Root systems are special sets of vectors which can generate reflection groups.
- A simple system can be thought of as a minimal generator for a root system.

Bibliography I


James E. Humphreys
Reflection groups and Coxeter groups.
Cambridge University Press, 1990.